

Customer Behavior Analysis using RFM Segmentation

Objective

The goal of this project is to analyze customer purchase behavior using transaction data and identify valuable customer segments. The analysis uses RFM (Recency, Frequency, Monetary) methodology to classify customers based on purchasing patterns.

Data Cleaning

The dataset was first cleaned to ensure accuracy and usability. Missing values in the Returns column were replaced with 0, duplicate records were removed, and the Purchase Date column was converted to datetime format.

Feature Engineering

Customer-level metrics were created from transaction data. These include:

- **Recency** – Days since the last purchase
- **Frequency** – Number of purchases made by the customer
- **Monetary** – Total amount spent by the customer

These metrics help evaluate customer engagement and purchasing value.

RFM Segmentation

Customers were segmented using RFM scores.

- High-Value Customers
- Mid-Value Customers
- Low-Value Customers

The segmentation helps businesses identify loyal customers and potential churn risks.

Visualization Insights

Several visualizations were created to analyze customer behavior.

Key findings:

- High-Value customers spend significantly more than other segments.
 - Mid-Value customers form the largest segment and contribute substantial revenue.
 - Low-Value customers have high recency values, indicating inactivity.
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Business Recommendations

1. Retain High-Value customers through loyalty programs and personalized offers.
 2. Encourage Mid-Value customers to increase purchase frequency.
 3. Re-engage Low-Value customers through discounts and marketing campaigns.
 4. Use customer segmentation to design targeted marketing strategies.
 5. Monitor recency to identify potential churn early.
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Conclusion

The RFM analysis successfully segmented customers based on purchase behavior. These insights help businesses improve customer retention, increase revenue, and design effective marketing strategies.